

Design intracranial stents with biodegradable materials that maintain structural integrity during deployment and initial healing period, then gradually degrade after their function is no longer needed

Executive Technology Brief

Domain: ml_data

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'physical_resistance_sticking' for intracranial stent: 12 MAUDE reports across 2 distinct devices (20200615 to 20230314); root-cause evidence from 0 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the r Unresolved gap: Across the 0 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, intervention, intracranial, introducing, maintain, malfunction, matching, modes, must, physical, problem, produ

Mechanism: Biodegradable sensing devices that can perform required functions within a programmed period and then disappear by safe resorption in the body or harmless degradation in the environment

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Design intracranial stents with biodegradable materials that maintain structural integrity during deployment and initial healing period, then gradually degrade after their function is no longer needed

Governing Model: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: CONDITIONAL; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: NOT ESTABLISHED.

ML-003: Empirical risk. Expression: $R_{emp} = \sum(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: loss.

ML-001: Expected utility of a decision rule. Expression: $EU = \sum_y P(y | x) * U(a(x), y)$. Applicability to this invention: APPLICABLE; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: U.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.